

Carlson Zheng

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EDUCATION

University of Alberta

BSc Computer Engineering Co-op

Edmonton, AB

2024 - 2029

SKILLS

Programming: Python, C/C++, Java, Bash, VHDL, Assembly

Frameworks/Libraries: Flask, OpenCV, Tensorflow, Pandas, numpy, mediapipe

Developer Tools: Git, STM32, Arduino, ESP32, KiCad, Linux, Visual Studio, Analog Discovery 3, SolidWorks

EXPERIENCE

Electrical Team Member

Dec 2025 - Present

Alberta Bionix

Edmonton, AB

- Designing a control system and integrating a complex sensor network for a lower-limb prosthetic to model a dynamic walking cycle using an STM32, moteous N1, rotary encoders, and IMUs.
- Interfaced rotary encoders and IMUs using embedded C, overcoming hardware abstraction challenges and leveraging hardware-accelerated data acquisition, reducing CPU load by 30%.
- Validated signal reliability across a network of 8+ interconnected sensors and components through 50+ hours of bench testing and hardware simulations.

Co-Founder/Co-President of Scarlett STEM

Oct 2022 - Jun 2024

Dr. E.P Scarlett High School

Calgary, AB

- Co-founded the school's first student-led STEM organization, scaling membership to 20+ students and securing resources to engage members in hands-on engineering and science related activities.
- Structured weekly coding sessions using Python and Java to mentor members in programming concepts such as object-oriented programming and data structures.
- Orchestrated weekly workshops and competitive STEM challenges, fostering a collaborative environment that increased student engagement by 50%.

Vice-President of Communications

Oct 2022 - April 2023

Munch-It / Junior Achievement

Calgary, AB

- Spearheaded a student-run company in the production of over 500+ units and achieving over \$1265+ in revenue and securing the Business Plan of the Year award out of the 30+ competing companies.
- Oversaw content development across social media and digital marketing platforms, while collaborating with the I.T. department to design and implement a website featuring intuitive and user-friendly interfaces.

PROJECTS

PomoRobo | *Git, C++, Arduino*

HackED - CompE Club, University of Alberta

- Engineered a 3-DOF robotic arm utilizing servo and stepper motors and ensuring <50ms latency between timer completion and mechanical response improving system functionality by 60%.
- Programmed a multi-state Pomodoro timer system using an Arduino and C++ that allows the arm to transition between idle, countdown, and active states, concluding with a synchronized servo wave.

Domain Detector | *Git, Python, Linux, OpenCV*

- Developed a real-time gesture recognition system in Python that accurately identifies specific hand signs by tracking 21 skeletal landmarks through a standard webcam.
- Trained a lightweight machine learning model using Tensorflow and Pandas that achieves over 93% accuracy and optimized to run smoothly on standard computer systems.

Discord Automation Bot | *Python, Flask, Render, Git, Discord*

- Created using Python and Discord API to automate 500+ monthly scheduled and location-triggered messages.
- Deployed to Render cloud hosting with a Flask 'heartbeat' monitoring system, ensuring 99.9% uptime.
- Implemented custom API endpoints for real-time management, reducing manual configuration time by 30% for pause/resume and location update features.